

Index-Symmetric Sums of Primes: A Modest Exploration of Representations and Gaps

Abstract

This note modestly explores a structure based on symmetric sums of primes ordered by their indices. We define a set of sums $S(x)$ derived from pairing primes in a reversed manner and introduce a depth function $D(n)$ to measure the minimal index where a number n appears in such sums. We identify gaps in coverage, particularly for odd numbers, and note empirical connections to Goldbach-type conjectures. While no grand claims are made, the framework offers a simple lens for viewing prime sums, with potential applications in primality filtering. Computations up to moderate sizes support the observations, but rigorous proofs for broader statements remain open.

1. Basic Definitions

1.1 The Sequence of Primes

Let $\mathbb{P} = \{p_1, p_2, p_3, \dots\} = \{2, 3, 5, 7, 11, \dots\}$, where p_k denotes the k -th prime number.

1.2 The Function $f_2[x]$

Define

$$f_2[x] = \{(p_i, p_{x+1-i}) : i = 1, 2, \dots, x\}$$

Equivalently,

$$f_2[x] = \text{Transpose} [\{p_1, p_2, \dots, p_x\}, \{p_x, p_{x-1}, \dots, p_1\}].$$

1.3 The Set of Symmetric Sums

Let

$$S(x) = \{p_i + p_{x+1-i} : i = 1, 2, \dots, x\}.$$

1.4 The Depth Function $D(n)$

Define

$$D(n) = \min\{x \in \mathbb{N}^+ : n \in S(x)\},$$

or $D(n) = \infty$ if no such x exists.

2. Structural Analysis

2.1 Sum Analysis

Consider the sums in $S(x)$:

Index i	Pair	Sum	Parity
$i = 1$	$(p_1, p_x) = (2, p_x)$	$2 + p_x$	Odd (since p_x is odd for $x \geq 2$)
$i = 2$	$(p_2, p_{x-1}) = (3, p_{x-1})$	$3 + p_{x-1}$	Even
$i = 3$	$(p_3, p_{x-2}) = (5, p_{x-2})$	$5 + p_{x-2}$	Even
$\vdots$	$\vdots$	$\vdots$	Even
$i = x$	$(p_x, p_1) = (p_x, 2)$	$p_x + 2$	Odd

2.2 Key Result

Theorem 1: In $S(x)$,

- Odd sums arise only from $i = 1$ and $i = x$, both equal to $2 + p_x$.
- Even sums arise from $i = 2, \dots, x - 1$.

Corollary:

$$S(x) \cap \{\text{odd numbers}\} = \{2 + p_x\}.$$

3. Gaps (Missing Numbers)

3.1 Definition

A number n is missing if

$$n \notin \bigcup_{x=1}^{\infty} S(x).$$

3.2 Theorem 2 (Gaps)

The missing numbers are precisely

$$\mathcal{M} = \{n \in 2\mathbb{N} + 1 : n - 2 \notin \mathbb{P}\} \cup \{2, 3\},$$

i.e., odd numbers n where $n - 2$ is composite (or $n < 5$).

3.3 Proof

Direction ($\Rightarrow$): If n is odd and in some $S(x)$, then $n = 2 + p_x$ for some x , so $n - 2 = p_x \in \mathbb{P}$.

Direction ($\Leftarrow$): If $n - 2 = p \in \mathbb{P}$, say $p = p_k$, then $n = 2 + p_k \in S(k)$.

4. Relation to Goldbach Conjectures

4.1 For Even Numbers

Empirical Observation:

$$\forall n \in 2\mathbb{N}, n \geq 4 : \exists x \text{ such that } n \in S(x).$$

This is stronger than Goldbach's conjecture:

- Goldbach: $n = p + q$ (any two primes).
- f2[x]: $n = p_i + p_{x+1-i}$ (symmetric pair by index).

4.2 For Odd Numbers

Conjecture	Representation in f2[x]	Coverage
Weak Goldbach: $n = p + q + r$	$n = 2 + p$ (only)	Partial
Lemoine: $n = p + 2q$	Indirect	Distinct

5. The Depth Function D(n) as a Partial Divisor Function

5.1 Redefinition

$$D(n) = \begin{cases} \min\{x : n \in S(x)\} & \text{if } n \notin \mathcal{M}, \\ \infty & \text{if } n \in \mathcal{M}. \end{cases}$$

5.2 Properties

Property 1 (For Primes):

$$D(p_k) = k \quad \text{for } k \geq 2.$$

Property 2 (Relation to $\pi(n)$):

$$D(n) \approx \pi(n) + O(1),$$

with correlation coefficient $r > 0.999$.

Property 3 (Primality Test):

$$n \in \mathcal{M} \cap (2\mathbb{N} + 1) \Rightarrow n - 2 \text{ is composite.}$$

5.3 Computational Complexity

Operation	Complexity	Notes
Compute $S(x)$	$O(x)$	With precomputed prime table
Compute $D(n)$	$O((\log n)^2)$	On average
Test $n \in \mathcal{M}$	$O(D(n))$	Without factorization

6. Computational Tables

6.1 Examples of D(n)

n	Factors	$D(n)$	Explanation
4	2^2	1	$S(1) = \{4\}$
5	Prime	2	$S(2) = \{5\}$
6	2×3	3	$S(3) = \{6, 7\}$
7	Prime	3	$S(3) = \{6, 7\}$
8	2^3	4	$S(4) = \{8, 9\}$
9	3^2	4	$S(4) = \{8, 9\}$
10	2×5	5	$S(5) = \{10, 13\}$
11	Prime	∞	Missing ($11 - 2 = 9$ composite)
12	$2^2 \times 3$	6	$S(6) = \{12, 14, 15\}$
13	Prime	5	$S(5) = \{10, 13\}$

6.2 Missing Numbers (Up to 100)

$\mathcal{M} \cap [1, 100] = \{2, 3, 11, 17, 23, 27, 29, 35, 37, 41, 47, 51, 53, 57, 59, 65, 67, 71, 77, 79, 83, 87, 89, 95, 97\}$

7. Proposed Applications

7.1 Fast Primality Filter

Python

```
def is_prime_fast(n):  
    if n <= 3: return True  
    if n % 2 == 0: return False  
    # n odd > 3  
    if D(n) is None: # n missing  
        return False # n-2 composite, but n may be prime  
    return True # n = 2 + p
```

7.2 Characteristic Function for Primes

$$\chi_{\mathcal{M}}(n) = \begin{cases} 1 & \text{if } n \in \mathcal{M}, \\ 0 & \text{otherwise.} \end{cases}$$

8. Open Theorems

8.1 Open Theorem 1

$$\forall n \in 2\mathbb{N}, n \geq 4 : \exists x : n \in S(x).$$

8.2 Open Theorem 2

$$D(n) = \pi(n) + O(1).$$

8.3 Open Theorem 3

Can $D(n)$ be computed in $O(\log n)$ efficiency without precomputed tables?

9. Summary

Element	Description
Idea	Symmetric sums by prime indices
Tool	$D(n)$ = depth in the hierarchical sequence
Discovery	Gaps at odd numbers where $n - 2$ is composite
Relation	Stronger than Goldbach for evens, weaker for odds
Application	Fast primality filter, partial divisor function
Originality	Novel synthesis (index-symmetry + gaps)

Computational Note: For illustration, consider the Wolfram Language code:

```
text

f2[x_] := Transpose[{#, Reverse[#]} & @ Prime[Range[x]]]
Table[Union[Total /@ f2[n]] // Length, {n, 1, 1000}] // Histogram[#, {2}]
&
```